Utilizing AI to Enhance Wind Energy Prediction: Integrated Research and Education

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Collaborative Research: REU Site: The Great Lakes Wind Energy Challenges

Motivation

Energy crisis & climate change → need → Sustainable energy solutions → through → Wind energy → challenge → Unsteady

Objective

- Identify **optimal locations** for wind energy projects
- Process **large** historical climate datasets to identify **patterns**
- Educational and outreach to enhance climate change **awareness**

Wind Data Analysis

Collect → Preprocess → Analyze → Visualize → Interpret

Wind Power Density [1]: $\mathcal{P}_w = \frac{1}{2}\rho U^3$ (1)

$\mathcal{P}_w$: wind power, U: wind speed, ρ: air density

Data Preprocessing

- raw data from Visual Crossing
- 10-minute intervals, *52,560 observations per year*
- linear interpolation for NaNs

Regional Analysis

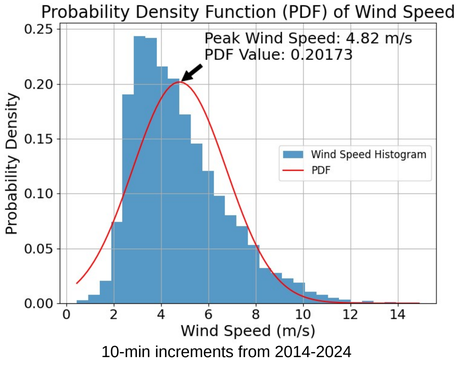
- 18 Locations were selected in Cleveland
- The north downtown region had the highest windspeed: 5.02 m/s
- The southeast region had the lowest windspeed: 3.78 m/s

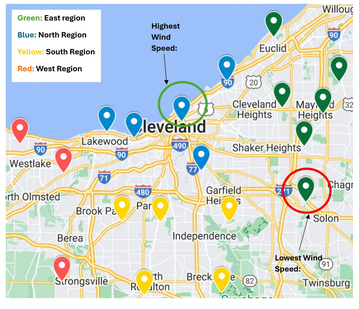
Probability Density Function (PDF) using Weibull distribution [1]:

$$f(u) = \frac{k}{c}\left(\frac{u}{c}\right)^{k-1} \exp\left(-(\frac{u}{c})^k\right), \text{ (k>0,c>0,u>0)} \quad (2)$$

Key Parameters:

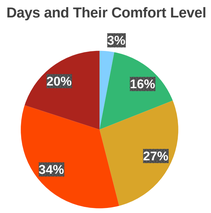
- Shape parameter (k): Influences the distribution's shape
- Scale parameter (c): Determines the distribution's shape





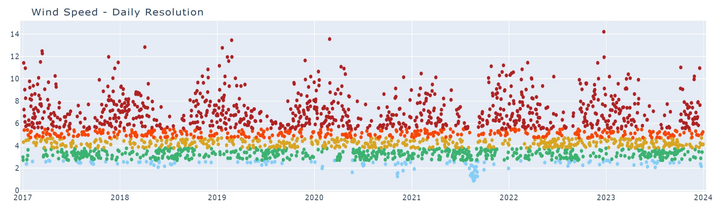
Interactive Dashboard

- Platform: Python
- Input: 7 years of climate variables
- Outcome: interactive user interface to visualize wind speed



Wind Speed Threshold and Corresponding Comfortable Activity	
Comfort level	Speed (m/s)
Sitting	<2.7
Standing	<3.8
Strolling	<4.7
Walking	<5.5
Uncomfortable	>5.5

DATA VISUALIZATION DASHBOARD



Artificial Neural Network

Date time	Wind Speed
1/1/2000 0:00	18.4
1/1/2000 0:10	
1/1/2000 0:20	
1/1/2000 0:30	
1/1/2000 0:40	
1/1/2000 0:50	12.9
1/1/2000 1:00	18.4
1/1/2000 1:10	
1/1/2000 1:20	
1/1/2000 1:30	
1/1/2000 1:40	
1/1/2000 1:50	14.7
1/1/2000 2:00	18.4

Convert Date Time to Integer Columns

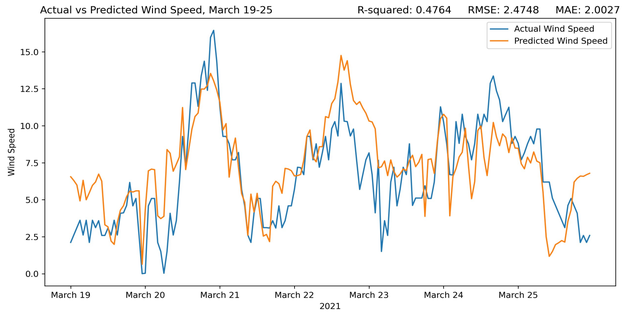
Convert Wind Speed to meters/second

year	day	time	Wind Speed
2000	0	0	5.1111111
2000	0	1	
2000	0	2	
2000	0	3	
2000	0	4	
2000	0	5	3.5833333
2000	0	6	5.1111111
2000	0	7	
2000	0	8	
2000	0	9	
2000	0	10	
2000	0	11	4.0833333
2000	0	12	5.1111111

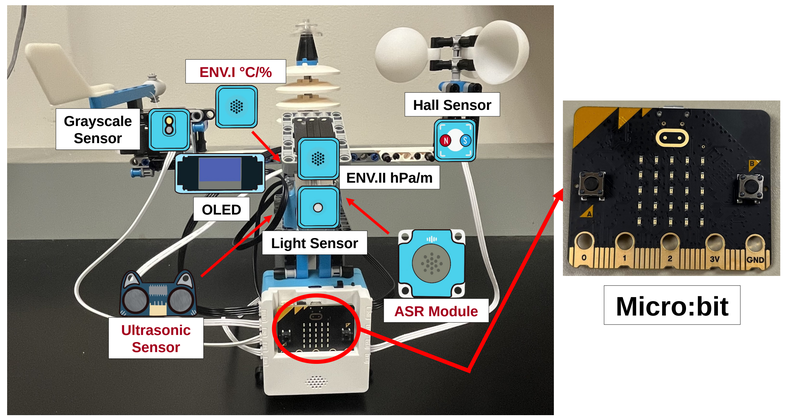
Fill empty rows using linear interpolation

year	day	time	Wind Speed
2000	0	0	5.1111111
2000	0	1	4.8055556
2000	0	2	4.5
2000	0	3	4.1944444
2000	0	4	3.8888889
2000	0	5	3.5833333
2000	0	6	5.1111111
2000	0	7	4.9055556
2000	0	8	4.7
2000	0	9	4.4944444
2000	0	10	4.2888889
2000	0	11	4.0833333
2000	0	12	5.1111111

- **4-layer** ANN with **64, 32, 16 neurons**, using **ReLU** activation.
- **12 input features:** year, day, time, temperature, dew point, wind direction, precipitation, wind gust, solar radiation, relative humidity, cloud cover, and sea level pressure
- **Trained** for **50 epochs** using **Adam optimizer**.
- Utilized **MSE loss function** for error minimization.



Research-Education Integration: Micro:bit Weather Station



- The **micro:bit** is a small, versatile microcontroller designed to introduce **programming** and **electronics** to beginners.
- Provides **hands-on learning** in programming and electronics

Educational Benefits

- Programming skills, electronics fundamentals, data collection, hands-on experience for a wide range of learners from K-12

Conclusion & Future Goal

- Interactive dashboard improves understanding of climate data for a wide range of users.
- ANN is a powerful regression model for wind speed prediction.
- Time-series models such as LSTM should be studied as a future research.
- Combined physics-based and data-driven approaches support informed decision-making for site selection process for wind energy projects.

References

[1] Goudarzi, N., Zhu, W. D., Delgado, R., & St. Pé, A. (2014). An Assessment on the Wind Energy Potential and Possible Solutions for Power Generation in Baltimore County in Maryland, USA. ASME Digital Collection, Paper No: FEDSM2014-21784, V01DT39A007.
[2] Reja, R.K., et al. "A New ANN Technique for Short-Term Wind Speed Prediction Based on SCADA System Data in Turkey." *Atmosphere*, vol. 14, no. 10, 2023, p. 1516, https://doi.org/10.3390/atmos14101516.
[3] Adamek, Kimberley, et al. "Pedestrian Level Wind Assessment through City Development: A Study of the Financial District in Toronto." *Sustainable Cities and Society*, 2017, http://dx.doi.org/10.1016/j.scs.2017.06.004.

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